Yachen Li

Earlier experience, up to 2024 | Website | GitHub

Education

Xi'an Jiaotong University **September 2019 – June 2024**

Bachelor of Clinical medicine *XI'AN, CHINA*

- Major Courses: Epidemiology, Biostatistics, Public health, Medical genetics, Immunology, Molecular Biology
- Awards: National first prize in the College Students' Innovation and Entrepreneurship Training Programme

Research Experience

Epigenetic Code Modeling Analysis Based on Stem Cell Lineage Trees **May 2022 – May 2023**

Team leader

- Utilized bioinformatics tools (e.g., STAR, rMATs, BWA, MACS2) to process and analyze RNA-seq and ChIP-seq data, focusing on stem cell lineage epigenetics.
- Employed R for interval data segmentation across five types of splicing data and constructed a splicing event-histone modification table.
- Won the first prize for the National Training Program and secured a patent on the 'Method for Identifying Key Regulatory Factors in the Determination of Cell Fate at Branch Points in Lineage Trees'.

Network Pharmacology **October 2022 – February 2023**

Core member

- Collected and processed traditional Chinese medicine data and disease targets utilizing R and Excel.
- Designed and constructed a comprehensive drug-active ingredient-target network, providing foundational support for advanced pharmacological studies.
- Conducted pathway analysis using Cytoscape and Metascape to elucidate intricate pharmacological interactions.
- Executed molecular docking techniques to predict and validate interactions between active ingredients and target proteins.

Work Experience

The First Affiliated Hospital of Xi'an Jiaotong University **June 2023 - June 2024**

Intern doctor

- Conducted daily patient consultations alongside senior doctors and maintained detailed medical records.
- Functioned as a Surgeon's Assistant during laparoscope operations and carried out precise wound suturing.
- Participated in weekly case discussions on patient scenarios and assisted with pre-operative preparations.

Shenzhen Bairen Technology Company **July 2022 - August 2022**

Bioinformatics Intern

- Employed techniques such as PERMANOVA for nuanced data interpretation and utilized LEfSe and DESeq for pinpoint biomarker analysis and differential abundance calculations in METH-induced CPP studies.
- Conducted analyses of within-group and between-group CPP variations, ensuring rigorous statistical validation to corroborate experimental conclusions.

Professional Experience

University Student Sleep Conditions and Influencing Factors Study **June 2022 – September 2022**

- Developed and administered a questionnaire to 81 students, processed data in Excel, and performed statistical analysis using SPSS.
- Collaborated with a supervisor to draft, discuss, and refine a comprehensive Chinese manuscript detailing the study's findings and implications.

The Basic Medical Center Project **September 2021 – May 2022**

- Conducted neurobiological experiments involving frog and mouse models to study muscular and systemic responses to hypoxia, CO poisoning, and sodium nitrite poisoning.
- Performed complex surgical procedures including the separation and cannulation of the rabbit's common carotid artery, along with the isolation of the vagus, sympathetic, and depressor nerves for comprehensive nerve studies.

Hospital Appointment System **June 2020**

- Designed a C#-based appointment system with LINQ to SQL integration.
- Crafted an intuitive GUI to facilitate user-friendly appointment bookings and management.

Leadership / Extracurricular

TLS Johnson & Johnson Club | Club President **September 2022 – Present**

- Elevated school's national ranking for Business Competition participation to a top 4 national ranking.
- Coordinated lectures and alumni engagement to assist students in finding mentors for career development and provide students with access to resume and application skills workshops.

Career Development Association | Club President **September 2022 – May 2023**

- Led 30+ alumni events, drawing 2,000+ students, thereby fostering alumni and student connections.
- Promoted gender equality through special Women's Day events.

Media Club | Club President **September 2021 – May 2022**

- Managed recruitment and marketing efforts for campus events, resulting in a significant increase in both membership numbers and 2,000 yuan in fundraising contributions to support club initiatives.

Volunteer work

- As part of the Rainbow Charity Society, mentored an underprivileged girl for two years, supporting her education through regular interactions and career coaching.
- Led a 22-member team to craft a comprehensive degree guidebook for incoming medical students and launched a related webpage with 2,300+ views.
- Assisted in health screenings at a local hospital, the 521 Community Hospital, recording vitals and monitoring blood pressure for over 800 seniors over a month.

Skills

Programming languages: R, Python, Bash, C#, Javascript
Bioinformatics tools: Trim Galore, BWA, STAR, rMATs, MACS2, SPSS
Computer skills: GitHub, WordPress, HTML, CSS, Linux